

Unit 13, Lesson 10: Equivalent Linear Expressions



Benchmark

MA.7.AR.1.2 Determine whether two linear expressions are equivalent.



Learning Target

- ☐ I can verify using substitution that linear expressions are equivalent.



13.10.1 Warm-Up: Origins of Algebra

The word **algebra** comes from the Arabic word الجبر, or al-jabr, which means, "the restoring of broken parts." The word algebra comes from the title of a book written by a Persian mathematician named Muhammad ibn Musa al-Khwarizmi in the 9th century.



1. Based on what you know about algebraic expressions and equations, explain why you think this meaning of the Arabic word makes sense to describe algebra.



13.10.2 Refresher: Evaluating Algebraic Expressions

Evaluate each expression for the given value of x .

1. $-3x - 8$ for $x = 5$

2. $4 - 2(x^2 - 7)$ for $x = -1$

3. $(4.5 + x) \cdot 3$ for $x = -\frac{5}{2}$



13.10.3 Collaboration: Solving Equivalent Expressions Using Substitution

1. With your partner, complete the table by choosing who will be Partner A or Partner B to determine if the expressions are equivalent.

Partner and Value	$8x + 4x - 2$	$2(6x - 1)$	Equivalent?
A $x = 2$			☐ yes ☐ no
B $x = 5$			☐ yes ☐ no
A $x = -5$			☐ yes ☐ no
B $x = (-2)$			☐ yes ☐ no

2. Complete the statement to describe the results in the last column of the table.

Depending on the value being substituted into the

expressions, the evaluated expressions

- always
- sometimes
- never

had the same value as each other.

3. What do the results tell you about the expressions $8x + 4x - 2$ and $2(6x - 1)$?

4. Rewrite $8x + 4x - 2$ by combining the like terms.

5. Use the distributive property to rewrite $2(6x - 1)$.

6. Discuss with your partner how the two expressions verify the results from your table.



13.10.4 Guided Instruction: Determining Equivalent Expressions Using Substitution

The two expressions $8x + 4x - 2$ and $2(6x - 1)$ are equivalent, but not equal. The expressions $12x - 2$ and $12x - 2$ are equal.

1. What is the difference between **equivalence** and **equality**?



Algebraic expressions are **equivalent** if they have different terms but are equal when evaluated using the same value for the variables. Algebraic expressions are **equal** if they have the same terms and operations.

2. Determine whether $5 - 3(2x + 1)$ and $4x + 2$ are equivalent using substitution of numerical values.

a. Complete the table.

x	$5 - 3(2x + 1)$	$4x + 2$
What is the value when $x = 0$?		
What is the value when $x = \frac{1}{2}$?		
Choose another value for x to substitute.		

b. Explain if the expressions are equivalent.

- c. Explain why the expressions were equal when "0" was substituted for x .

- d. Discuss with your partner the advantages and disadvantages of using "0" to substitute for a variable to check for equivalence.

3. Determine whether $3 + x(4 - 2)$ and $x(3 - 7) + 5$ are equivalent using substitution of numerical values.

a. Complete the table.

x	$3 + x(4 - 2)$	$x(3 - 7) + 5$
What is the value when $x = 1$?		
What is the value when $x = 0$?		
Choose another value for x to substitute.		

b. How many values are best to use when substituting values in for the variable to check for equivalency?



13.10.5 Guided Practice: Equivalent Expressions Using Substitution

1. Consider the three expressions below. Determine whether the expressions are equivalent by substituting the given value of x .

a. Complete the table.

x	$4x - \frac{1}{2}(6 + 8x)$	$8x - 3$	$-(4x + 3) + 4x$
$x = 3$			
$x = -2.75$			
$x = \underline{\hspace{2cm}}$			

- b. What conclusion can you draw about the expressions from the information in the table?

2. Determine whether the expressions $\frac{2}{5}(15x - 30y) + 10x$ and $4(4x - 2y)$ are equivalent.

x and y	$\frac{2}{5}(15x - 30y) + 10x$	$4(4x - 2y)$
$x = -2$ $y = 0$		
$x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$		



13.10.6 Your Turn!

1. Determine which expressions are equivalent by substituting the same values for the variable.

A $(5x - 3) - \frac{3}{4}(8 + 2x)$	B $-5\frac{1}{4}(4x - 1)$
C $9 - 6\left(\frac{-2}{3}x + 4\right)$	D $\frac{3}{2}x - 9 + 2x$
E $\left(\frac{30}{2}\right)(-1) + 4x$	F $6 - 12\frac{1}{2}x - \frac{3}{4} - 8.5x$

2. Complete each statement using the expressions from the table.

_____ is equivalent to _____

_____ is equivalent to _____

_____ is equivalent to _____